

Recommended Indicators per Slot, by Asset Class

What this is: a single best-pick per category slot (Trend / Momentum / Volatility / Volume) for each asset class, combining (a) the empirical flat IC screen (`grouped_ic_results.csv`) and (b) domain intuition where the data is silent or thin. The goal is to pick the *best available option per slot* even when the edge is weak.

Honesty header — read once: - Empirical ICs here are small (mostly $|IC|$ 0.03–0.06). Surviving the screen means *not obviously useless*, not *validated* or *tradeable after costs*. - The **only strong, robust empirical edge** is **FX short-horizon mean-reversion** ($|IC| \approx 0.20\text{--}0.27$). Everything else is "best-in-a-weak-field." - **Sign matters.** NEG = contrarian (fade the reading); POS = directional (follow it). - Rows marked (*intuition*) are domain reasoning, **not** backtested here — used only where the flat screen found no FDR survivor (notably Metals/Energy Trend & Momentum). - Regime-conditioned results were **artifact-dominated** (overlap/small-sample), so they are used only as soft hypotheses, never as the basis for a pick.

What each slot is (and why the architecture has four)

The pipeline fills four slots with exactly one indicator each. They answer different, non-overlapping questions. You do **not** ask "does RSI beat ATR" — they live in different slots and do different jobs.

1. Trend — "Which way, and is there a trend at all?"

Measures **direction and trend strength**. A trend indicator's job is to tell you whether price is in a persistent directional move (follow it) or ranging (ignore it / fade it). Trend indicators are **laggy by construction** — they trade timeliness for robustness. In this study trend signals were the weakest slot on flat history, because "trend" only pays in specific regimes and averages to ~0 across all history.

2. Momentum — "How stretched is the current move?"

Measures the **rate of change / overbought-oversold state**. Bounded oscillators (RSI, Stoch, UO) are really **mean-reversion** tools at short horizons — a high reading means "stretched, likely to snap back," not "strong, keep going." Raw momentum (MOM, ROC) is the opposite: it measures persistence. That is why momentum-slot signs flip between contrarian (oscillators) and directional (raw momentum) depending on asset and horizon.

3. Volatility — "How big are the moves right now?"

Measures **magnitude, not direction**. This is the one slot that "works" almost everywhere — but only because **volatility is autocorrelated** (vol clusters: calm follows calm, storms follow storms). It does **not** predict return *direction*. Its real job is **position sizing, stop placement, and regime gating** — turn other signals off when vol is extreme, size down when NATR spikes.

4. Volume — "Is the move backed by participation?"

Measures **conviction / money flow**. Volume indicators confirm or diverge from price: price up on rising ADOSC/CMF = real buying; price up on falling volume = suspect. **Not available for spot FX** (no centralized volume). Weakest and noisiest slot overall; best used as a *confirmation filter*, not a standalone signal.

Master table — best pick per slot per asset class

Legend: **sign** NEG=contrarian / POS=directional · **h** = horizon (trading days) where the edge is clearest · (*intuition*) = domain pick, not from the screen.

Asset class	Trend	Momentum	Volatility	Volume
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Equity indices	VORTEX-14 · NEG · h1	TSI-13-25 · NEG · h1	Ulcer Index-14 · POS · h1	ADOSC-3-10 · NEG · h1
FX (majors)	VORTEX-14 · POS · h1	Ultimate Osc 7-14-28 · NEG · h1 ■	RVI-14 · NEG · h1	— (no volume)
Rates (yields/USTs)	Aroon Osc-14 · POS · h10	MOM-10 · NEG · h1	RVI-14 / ATR-ratio · h1	ADOSC-3-10 · NEG · h1
Credit (IG/HY)	VORTEX-14 · NEG · h20	TSI-13-25 · NEG · h10	Ulcer Index-14 · POS · h1	CMF-20 · NEG · h5
Metals (gold/silver)	ADX-14 gate + EMA50/200 (intuition)	ROC-10 / MOM-10 (intuition)	ATR-ratio-14 · POS · h1	CMF-20 · NEG · h1
Energy (WTI)	Donchian-20 breakout (intuition)	MOM-10 directional (intuition)	ATR-ratio-14 · POS · h10	ADOSC (intuition)

Discrete cross/threshold signals that survived (best per class, for regime/state gating):

Asset class	Trend (discrete)	Momentum (discrete)
Equity	EMA 50/200 cross (golden/death), spread +0.017 @h20	RSI 30/70 threshold, +0.021 @h20
Rates	Supertrend 10-3, +0.030 @h20	RSI threshold, -0.067 @h20 (contrarian)
Credit	PSAR cross	RSI threshold, +0.018 @h20
Metals	PSAR cross	RSI threshold, +0.015 @h20
Energy	Supertrend 10-3	RSI threshold, -0.044 @h20
FX	EMA 50/200 cross	RSI threshold, +0.019 @h20

Per-asset-class detail: signals, technical strategies, subjective strategies

For each class: **Empirical read** (what the data says) → **Technical strategy** (mechanical rules) → **Subjective strategy** (discretionary/domain overlay).

Equity indices (^GSPC, ^NDX, ^STOXX50E, ^FTSE, ^GDAXI, ^N225, ^KS11, ^HSI)

Empirical read. Everything wins at **h=1** and is **contrarian** (Vortex, TSI, ADOSC all NEG) except volatility (Ulcer Index, POS). Translation: at the 1-day horizon, equity indices **mean-revert** — stretched moves snap back. Volatility is the sturdiest slot (vol clustering). Magnitudes are small (|IC|~0.03–0.045).

- **Trend — Vortex-14.** Signal: VTXP crosses above VTXM = uptrend, below = downtrend. Here it reads *contrarian at h1* (extended trend readings precede a snap-back).
- **Momentum — TSI-13-25.** Double-smoothed momentum; zero-line and signal-line cross. Contrarian at h1.
- **Volatility — Ulcer Index-14.** Downside-only volatility (depth × duration of drawdown). Rising UI = accumulating stress. POS IC = high stress precedes higher short-horizon forward returns (the "buy-the-dip / vol-mean-reversion" effect).
- **Volume — ADOSC-3-10 (Chaikin oscillator).** Volume-momentum; contrarian at h1.

Technical strategy (mechanical). - *Mean-reversion core:* on a 1–5 day horizon, **fade** RSI<30 (long) / RSI>70 (short) **only when EMA50>EMA200** (long side) — i.e. buy dips in uptrends, the golden-cross regime filter that survived at +0.017 spread. - *Vol gate:* compute Ulcer Index / NATR percentile. **Size up** mean-reversion entries when UI is elevated (dips in stress get bought); **stand down** entirely when NATR is in its top 5% (crash risk — reversion breaks). - *Trend overlay:* use EMA50/200 as the *only* trend gate; don't trade counter-trend mean-reversion when the 200-day slope is negative.

Subjective strategy (discretionary). - Equity indices have a **long-side drift** (equity risk premium). Bias mean-reversion longs over shorts; treat short signals as "reduce/hedge," not "go net short." - Short-term reversion **inverts into momentum around macro catalysts** (CPI, FOMC, earnings season). Suspend "buy the dip" into a known event; the dip may be information. - Cross-index divergence is a tell: if ^NDX cracks while ^GSPC holds, trust the weaker one (breadth deteriorating). Use ^N225/^HSI overnight as a sentiment read for the US open.

FX majors (DX-Y.NYB, EURUSD=X, USD/CHF) ■ the one real edge

Empirical read. Ultimate Oscillator is strongly contrarian at h1 (|IC|≈0.22) — the standout finding of the whole study, and it *sharpened* inside calm-bull regimes. FX majors **mean-revert hard at 1–5 days**. Trend (Vortex) is weakly directional; volume slot is **not applicable** (spot FX has no consolidated volume).

- **Momentum — Ultimate Oscillator 7-14-28.** Multi-timeframe oscillator. $UO > 70$ = fade long / short bias; $UO < 30$ = fade short / long bias. This is the pick to actually use.
- **Trend — Vortex-14** (directional, weak) as a *filter* only.
- **Volatility — RVI-14** for sizing.

Technical strategy (mechanical). - *Reversion engine*: short EUR/USD (or long USD) when **UO > 70**, cover/flip when UO crosses back below 50; symmetric on the downside. Best at h1–h5, then decays. - *Confirm with RSI band*: require $RSI(14) > 70$ or < 30 alongside UO for higher-conviction entries (the RSI-threshold discrete also survived, +0.019 spread). - *Regime sharpener*: the edge is strongest in **calm-bull (RC-01)**; widen bands or stand aside when VIX-Z is elevated (shock breaks mean-reversion → momentum).

Subjective strategy (discretionary). - Respect the **carry/policy backdrop**: FX mean-reversion works within a range but *fails during trends driven by rate-differential shifts* (e.g. a hiking cycle). Check the 2Y yield spread direction before fading a move. - **DXY as the hub**: trade EUR/USD and USD/CHF as expressions of USD strength; if DXY is breaking a multi-month level, drop the contrarian bias and go with the break. - Time-of-day/liquidity matters (London/NY overlap); reversion signals on thin Asian liquidity are less reliable — discount them.

Rates (^TNX yield, TLT)

Empirical read. ^TNX is **yield-based** (signals predict yield *changes*, not price). Trend (Aroon Osc) is **directional at h10** — rate trends persist over ~2 weeks; short horizon (MOM, RVI, ADOSC) is contrarian. The RSI-threshold discrete is notably contrarian (−0.067 @h20).

Technical strategy. - *Trend-follow the medium horizon*: Aroon Oscillator > 0 (and rising) = yields trending up → short duration / short TLT; use a 10-day holding horizon. - *Fade the extreme*: at h1, MOM/RSI extremes in yields mean-revert — fade a 1-day yield spike. Supertrend(10,3) flip (+0.030 @h20) as the trend-state gate.

Subjective strategy. - Rates are **policy-driven**: the dominant variable is the Fed path, not the chart. Use technicals for *timing within* a macro thesis, never against it. - Watch the **2s10s curve**: a bull-steepener vs bear-flattener changes which end of the curve leads. TLT (long end) and ^TNX (10Y) should move inversely in price terms — divergence flags a curve move worth trading on the curve, not outright.

Credit (LQD, HYG)

Empirical read. The **best non-FX magnitudes** (Vortex −0.10 @h20, TSI −0.09 @h10) — both **contrarian**. Credit spreads are highly autocorrelated and mean-revert; these ETFs have real volume so CMF (money flow) is meaningful (−0.07 @h5).

Technical strategy. - *Spread reversion*: fade extended moves in HYG at the 1–2 week horizon (Vortex/TSI contrarian); confirm with CMF — a price drop on *outflows* ($CMF < 0$) is more likely to extend, a drop on *inflows* is a buyable reversion. - *Vol gate*: Ulcer Index on HYG as a stress meter — rising UI = risk-off; halt reversion longs.

Subjective strategy. - Credit is the **canary**: HYG weakness *leading* equities is a genuine risk-off tell — trust it over the chart. Treat HYG less as a tradeable and more as a **regime input** for the equity book. - Liquidity in HY dries up in stress;

technical signals get unreliable exactly when you most want them. Size accordingly.

Metals (GC=F gold, SI=F silver)

Empirical read. No Trend or Momentum FDR survivor — the flat screen found no standalone directional edge (they wash out across regimes). Only Volatility (ATR-ratio, POS) and Volume (CMF, contrarian) survived. This is the clearest **regime-masking** candidate: gold's momentum likely lives in risk-off / real-rate regimes and averages to zero flat.

- Volatility — **ATR-ratio-14** for sizing (works).
- Trend/Momentum — (*intuition*) see below; the screen is silent, not disproving.

Technical strategy. - *Volatility-scaled trend (intuition)*: gold **trends** structurally. Use **ADX-14 as a gate** (only take trend trades when ADX>25) plus an **EMA50/200** direction filter; size inversely to ATR-ratio. This is the classic metals approach even though flat IC is ~0 — because the payoff is regime-concentrated. - *Silver = high-beta gold*: same signals, smaller size, wider stops (ATR-ratio is structurally higher).

Subjective strategy. - Gold is a **real-rates and USD** instrument first, a chart second. Falling real yields + weak USD = the tailwind regime; that's when to lean into trend signals. - Treat gold as **portfolio insurance**: its best trends coincide with equity stress, so a metals long is partly a hedge — hold through chop you'd cut elsewhere.

Energy (CL=F WTI)

Empirical read. Like metals: **no Trend/Momentum survivor**, only **Volatility (ATR-ratio, POS, |IC|≈0.08 @h10)** — the strongest volatility read in the study. WTI is a **continuous front-month stitch** (roll artifacts at short horizons — discount h1).

Technical strategy. - *Breakout trend-follow (intuition)*: commodities are the classic CTA/trend market. **Donchian-20 channel breakout** entries, ATR-based trailing stop, is the archetypal crude system — flat IC is ~0 because trends are regime-bound (supply shocks), but the fat-tailed payoff favors trend-following. - *Vol regime is real*: ATR-ratio elevated → widen stops and cut size; crude vol clusters violently.

Subjective strategy. - Crude is driven by **inventories, OPEC, and the futures curve** (backwardation vs contango). Backwardation = structural tailwind for longs (positive roll); flip bias with the curve, not just the chart. - Geopolitical spikes are **mean-reverting fear** unless supply is actually removed — fade the panic wick, hold the structural break.

How to assemble the four slots into one decision

The slots combine, they don't compete: 1. **Volatility** sets the environment → position size and whether to trade at all (stand down at vol extremes). 2. **Trend** sets the *direction filter* → only take signals aligned with it (or only fade when ranging). 3. **Momentum** sets the *entry timing* → the oscillator extreme is your trigger. 4. **Volume** is the *confirmation* → veto entries where flow diverges from price.

Default per-class posture from this study: - **FX**: momentum-led **mean-reversion** (UO), vol-gated. (*the real edge*) - **Equity**: dip-buying **mean-reversion** in up-trends, vol-gated, event-aware. - **Credit**: spread **mean-reversion** + use as a risk-off regime input. - **Rates**: medium-horizon **trend-follow** the yield, fade 1-day extremes. - **Metals/Energy**: **volatility for sizing is the only proven piece**; trend-follow is the intuitive best-guess for direction, to be validated by regime-conditioned testing on the right (risk-off / supply-shock) cells — not by flat IC.

Caveats (do not skip)

- Empirical picks come from a **flat, full-history, single-indicator Spearman IC screen**. Not walk-forward, not cost-adjusted, not out-of-sample.
- (*intuition*) rows and every "Subjective strategy" block are **domain reasoning, not results** — they are hypotheses for the regime-conditioned run to test on single stocks and the right regime cells.

- Metals/Energy Trend & Momentum picks are explicitly **not supported by the flat data** (no FDR survivor) — included because the goal was "best available option," and their absence flat is itself the regime-masking signal to chase next.
- Signs (contrarian vs directional) are as important as the indicator name. Re-check them per asset before deploying.